

DNS over Avian Carriers (DoAC)

draft-cruzgonzalez-ipoac-dns

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STATUS

Active individual Internet-Draft — open for review and comments

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1. WHAT AN APRIL 1ST RFC IS — AND WHY IT MATTERS

RFCs (Requests for Comments) are the technical documents that define how the internet works: from how email is delivered to how your browser negotiates a secure connection. They are published by the IETF (Internet Engineering Task Force), the international body responsible for standardizing internet protocols.

Since 1989, the IETF has maintained a tradition: publishing a humorous RFC every April 1st. The most famous is RFC 1149 (1990), which specifies the transmission of IP datagrams via homing pigeons. In 2001, a group of Norwegian engineers implemented it in practice, achieving an average ping time of 3,000 seconds and 55% packet loss — results considered acceptable given the circumstances.

These documents are no less rigorous than any other RFC. They follow the same technical format, use the standard normative language (MUST, SHOULD, MAY), are reviewed by the internet engineering community, and are permanently registered in the IETF archive. The difference is that their stated purpose is humorous — but their technical execution is entirely serious.

Contributing to this tradition is a form of recognition within the global technical community: it requires deep knowledge of internet standards, the ability to identify a real (if absurd) technical gap, and the skill to write in the precise register of a normative document. Not everyone gets to publish one.

2. THE PROBLEM — THREE DECADES OF AN UNACCEPTABLE GAP

RFC 1149 solved the transport layer. RFC 2549 added Quality of Service. RFC 6214 extended support to IPv6. For over thirty years, the avian-carrier protocol stack advanced with rigor and determination.

And yet, no one had addressed the most basic question: how does a host on an avian-carrier network resolve a hostname?

The only available solution to date was to hardcode the IP address directly onto the message attached to the carrier. This approach, as any engineer with judgment will recognize, does not scale. It does not support load balancing across multiple lofts. And it makes pigeon renumbering events what the draft's author describes, with admirable restraint, as "catastrophically disruptive".

3. THE PROPOSAL — DOAC CLOSES THE GAP

DNS over Avian Carriers (DoAC) specifies a complete mechanism for resolving domain names over avian-carrier infrastructure. DNS queries are transcribed onto microfilm — recommended over plain paper in regions with annual precipitation exceeding 50mm — and attached to the carrier's left leg. Responses go on the right. Implementations MUST NOT attach both a query and a response to the same leg simultaneously.

The document introduces the AA (Avian Address) DNS resource record type, which encodes the geographic coordinates of the destination loft rather than a conventional IP address. The TTL (Time To Live) field retains its standard DNS semantic meaning, though it acquires an additional biological interpretation: operators SHOULD NOT configure TTL values exceeding the expected lifespan of the carrier. Values above 15 years are NOT RECOMMENDED.

For situations in which no available carrier knows the route to the destination, the document defines the Pigeon of Last Resort (PoLR): a pre-trained carrier equivalent to the recursive resolver of last resort. The condition in which a carrier fails to arrive at its destination within the expected window is designated NXPIGEON.

The Trusted Courier Pigeon (TCP) is the carrier designated for the transport of sensitive keying material. The acronym collision with Transmission Control Protocol, the document notes, "is noted and considered appropriate".

4. SECURITY CONSIDERATIONS — A THOROUGH ANALYSIS

The document gives rigorous attention to the security threats inherent to this architecture. Among those identified: the Hawk-in-the-Middle (HitM) attack, in which an intermediate agent — specifically, a bird of prey — intercepts the carrier before it reaches its destination; pigeon spoofing via Plumage-Based Authentication; and Loft Hijacking.

Also documented is the Covert Feather Channel: a side channel through which sensitive information may be exfiltrated by analyzing the wear patterns of the carrier's plumage. Denial of Flight (DoF) and routing anomalies caused by migration season complete the threat model.

The hungry cat is formally classified as a physical layer threat.

For large DNS responses — such as DNSKEY RRsets — the document recommends using a larger bird. The use of eagles is OPTIONAL, though expressly acknowledged as impressive.

5. STATUS AND NEXT STEPS

The draft is currently under review with the IETF as an individual Internet-Draft. The document remains open for comments from the technical community. The full text is available in English on the IETF Datatracker.

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This document is a press summary. The original Internet-Draft, in English, is available on the IETF Datatracker and constitutes the authoritative technical reference. DNS over Avian Carriers is a humorous document in the tradition of the IETF's April 1st RFCs. The author assumes full technical responsibility for its contents.